

## Research question and claim or hypothesis

**Research Question:** How does the choice of flow-matching prior (Gaussian white noise vs Gaussian-process prior) interact with the velocity network's architectural capacity (unstructured AdaLN MLP vs temporally structured S4)?

**Claim/Hypothesis:** Regressing a matched-GP velocity field using an unstructured MLP (like Sundial's TimeFlow head) in raw observation coordinates is poorly preconditioned. Common-mode gradients dominate, causing residual coordinates to jitter. Conversely, a temporally structured head (S4, like TSFlow) natively handles this geometry. The issue lies in the Jacobian of the head, not identifiability.

## Task, horizon or decision setting, and intended output

**Task:** Independent-coupling conditional flow matching for time series on raw observation coordinates (RF).

**Setting:** A fixed small patch size/horizon (e.g.,  $F=16$ ) conditioned on a small lookback hidden state.

**Intended Output:** A 2x2 empirical evaluation (Source×Head) proving whether a structured observation-space source requires a temporally structured velocity net. The output will be a table of velocity errors and generated patch plots, leading to a specific design constraint recommendation.

## Data source, access, licensing, and limitations

**Data Source:** Strictly synthetic data for Tasks 1 and 2. Univariate lookback and horizon patches sampled from a Gaussian process (GP) with a long lengthscale. Optional Task 3 uses a real series that is naturally smooth (e.g., a long-lengthscale GIFT-Eval split or sine-plus-slow-trend).

**Access & Licensing:** Synthetic data requires no license. GIFT-Eval is open source (MIT/Apache).

**Limitations:** The synthetic GP represents a narrow, near-collinear geometric configuration. The project does not claim that Sundial fails on general volatile data like TimeBench; it isolates and tests the smooth/structured source behaviour.

## Train–validation–test protocol and leakage audit

**Protocol:** Datasets will be strictly split into independent synthetic draws for train, validation, and test sets. Two source families (isotropic Gaussian white noise and matched-GP draw) will be drawn independently of the target under standard conditional-flow-matching coupling.

**Leakage Audit:** Since the primary data is synthetic, data leakage is trivially avoided by using different random seeds to generate the test set. For the optional real dataset, strictly chronological splits will be used with no overlap between lookback and horizon windows.

## Metrics and a short justification

**Spectrum Energy (Task 1):** Fraction of variance in the first two principal components (PCs) of the target and velocity. **Justification:** Validates the minimum passing score—ensures the synthetic GP is actually smooth and near-collinear.

**Common-mode velocity error:** MSE of the velocity projected onto the leading PC of  $(y-\epsilon)$ . **Justification:** Measures if the head captures the primary trend.

**Residual-PC velocity error:** MSE on the orthogonal complement (PCs 3...F).

**Justification:** Core metric for the claim; checks if the model jitters on higher-frequency/residual structures.

**Sample-path roughness:** Mean adjacent jump ( $|x_{i+1} - x_i|$ ) of generated patches after Euler integration.

**Justification:** Measures if the generated output matches the physical smoothness of the GP targets.

### **Baseline ladder (naive or classical → modern)**

**Naive/Published Baseline:** Unstructured AdaLN MLP with an isotropic Gaussian white-noise source (Sundial's standard TimeFlow head setting).

**Intermediate:** S4 head with an isotropic Gaussian white-noise source.

**Modern/Stress-Test:** S4 head with a Gaussian-process source (TSFlow-style velocity net).

**Alternative/Control:** PCA-whitened MLP or first-two-PC map with a GP source.

### **Method and the smallest decisive experiment**

**Method:** Train tiny velocity nets of comparable width, conditioned on a small lookback encoder (not a foundation backbone).

**Smallest Decisive Experiment (Task 2):** A 2x2 grid crossing Sources (White Noise vs. Matched-GP) with Heads (AdaLN MLP vs. S4). If the AdaLN MLP matches the common mode but loses on residual PCs and roughness in the GP→GP cell, while S4 succeeds, the core claim is proven. If AdaLN MLP matches S4 across all metrics, the prior rank-deficiency hypothesis is falsified.

### **Planned ablations or robustness checks**

**Head/Coordinate Ablation (Task 3):** Keep the matched-GP source but replace the head with a PCA-whitened MLP or a degree-1 polynomial readout to test if the extra F-2 channels were just unused and causing the gap.

**Robustness Check:** Evaluate the GP→GP cell on a real smooth time series to ensure the findings aren't an artefact of the specific synthetic GP generator (provided it passes the Task 1 spectrum test).

### **Compute budget, runtime, and fallback**

**Compute budget & Runtime:** The experiments use tiny velocity nets on small patches (F=16) without pretraining on TimeBench. A single standard GPU (e.g., T4/L4) or even a modern CPU will run the full 2x2 grid in under a few hours.

**Fallback:** If the S4 layer implementation proves too complex or unstable to train quickly, we will fall back to using a standard causal 1D-CNN (WaveNet/DiffWave style) as the temporally structured head proxy, or focus entirely on the PCA-whitened MLP control.

### **Roles, repository, and weekly plan**

Roles:

**Egor Serov (Contact Person):** Repository management, AdaLN MLP implementation (Task 2), training loops for the 2x2 grid.

**Frolov Kirill:** Synthetic data generation (Task 1), PCA spectrum tests, and lookback encoder implementation.

**Daniil Koblov:** S4 model implementation (Task 2), computation of evaluation metrics (common-mode/residual errors, sample-path roughness).

**Vasilii Lyamin:** Task 3 ablations (PCA-whitened control head, optional real series check), plotting/visualizations, and final report writing.

**Repository:** [https://github.com/egorser0v/FM\\_stress](https://github.com/egorser0v/FM_stress)

### **Weekly Plan:**

**Week 1:** Repo setup. Task 1: Generate synthetic GP data and compute PCA spectrum tests.

**Week 2:** Task 2: Implement lookback encoder, AdaLN MLP, and S4 models. Run the 2x2 grid experiments.

**Week 3:** Compute common-mode/residual errors and sample-path roughness. Task 3: Implement the PCA-whitened control head.

**Week 4:** Final analysis, visualisation of generated patches vs targets, drawing conclusions, and writing the recommendation paragraph.

### **Main scientific and execution risks**

**Scientific Risk:** The synthetic generation might yield velocities that are completely full rank (kernel not smooth enough on  $F=16$ ) or totally collapsed to constants after instance norm. Mitigation: Explicit spectrum check in Task 1 before moving to Task 2; adjust lengthscale or patch length accordingly.

**Execution Risk:** S4 layers can be tricky to initialise and tune. Mitigation: Use well-tested open-source implementations (e.g., from state-spaces/s4 or TSFlow repo) and keep the model exceptionally small.

### **Intended AI use and how outputs will be checked**

**Intended AI Use:** LLMs (like ChatGPT/Claude) may be used to draft standard PyTorch boilerplate code, assist with the S4 module implementation, and polish the final report text.

**Verification:** All AI-generated code will be strictly unit-tested. Specifically, we will validate the metrics computation (MSE projections and roughness) on dummy tensors with known values. Data geometries will be verified via the Task 1 spectrum test before any training begins.